

CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 9 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. In a single knock-out tournament consisting of 25 teams, exactly how many matches will be played in total to determine the winner? (A) 25 (B) 24 (C) 12 (D) 26 *Unit 1: Management of Sporting Events | [PYQ] | Tag: Very High***

Q2. The postural deformity characterized by the collapsing of the inner arches of the feet, resulting in the entire sole touching the ground, is medically termed as: (A) Pes Planus (B) Genu Valgum (C) Genu Varum (D) Kyphosis *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***

Q3. Which of the following yoga asanas translates to the "Camel Pose" and is highly beneficial in correcting postural deformities of the spine? (A) Vajrasana (B) Ushtrasana (C) Halasana (D) Tadasana *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [Practice] | Tag: High***

Q4. "Spirit in Motion" is the officially recognized motto of which international sporting organization? (A) Special Olympics (B) Deaflympics (C) Paralympic Games (D) Commonwealth Games *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: Very High***

Q5. Which essential fat-soluble vitamin is required by the human body to effectively absorb calcium from the diet, thereby promoting healthy bone density? (A) Vitamin A (B) Vitamin B (C) Vitamin C (D) Vitamin D *Unit 5: Sports & Nutrition | [PYQ] | Tag: Very High***

Q6. The 'Johnson-Methney Test' consisting of items like the front roll and jumping half-turn is specifically designed to measure an individual's: (A) Cardiovascular endurance (B) Motor educability (C) Lower body strength (D) Agility *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: High***

Q7. The maximum volume of oxygen that can be absorbed, transported, and consumed by the working muscles per minute during intense aerobic exercise is known as: (A) Oxygen Intake (B) Tidal Volume (C) Maximum Oxygen Uptake ($\dot{V}O_2$ max) (D) Cardiac Output *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

Q8. According to the principles of biomechanics, the force of friction always acts in the __ direction to the intended motion of an object. (A) Same (B) Opposite (C) Perpendicular (D) Upward *Unit 8: Biomechanics and Sports | [PYQ] | Tag: Very High***

Q9. The 'Tabular Method' is a systematic mathematical procedure used to draw fixtures for which specific type of tournament? (A) Knock-out Tournament (B) League Tournament (C) Consolation Tournament (D) Challenge Tournament *Unit 1: Management of Sporting Events | [Practice] | Tag: Very High***

Q10. A female athlete experiencing irregular or infrequent menstrual cycles occurring at intervals longer than 35 days is medically diagnosed with: (A) Amenorrhea (B) Menarche (C) Oligomenorrhea (D) Osteoporosis *Unit 2: Children & Women in Sports | [PYQ] | Tag: Very High***

Q11. Which specific *Pranayama* involves controlled inhalation through the left nostril and exhalation through the right nostril, commonly known as alternate nostril breathing? (A) Kapalbhati (B) Nadi Shodhana (C) Suryabhedan (D) Sitli *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: High***

Q12. In the Special Olympics, the equitable process of grouping athletes into competitive brackets based on their age, gender, and primarily their functional sports skill level is termed as: (A) Classification (B) Categorization (C) Divisioning (D) Seeding *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: Very High***

Q13. Which of the following nutritional components is a classic example of a 'Complex Carbohydrate'? (A) Glucose (B) Fructose (C) Sucrose (D) Starch *Unit 5: Sports & Nutrition | [Practice] | Tag: High***

Q14. In the Rikli & Jones Senior Citizen Fitness Test battery, what is the standardized weight of the dumbbell used for women in the 'Arm Curl Test'? (A) 4 lbs (B) 5 lbs (C) 8 lbs (D) 10 lbs *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q15. An acute sports injury in which a bone is completely forcefully displaced from its articulating joint surface is known as a: (A) Fracture (B) Contusion (C) Sprain (D) Dislocation *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

Q16. Assertion (A): Newton's Second Law of Motion is also universally known as the Law of Momentum. **Reason (R):** The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports | [PYQ] | Tag: Very High***

Q17. Assertion (A): 'Seeding' is a tournament procedure by which strong teams are strategically placed in fixtures so that they do not meet each other in the very first round. **Reason (R):** This procedure guarantees that top teams are protected from early elimination, maintaining spectator interest for the final rounds. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 1: Management of Sporting Events | [Practice] | Tag: Very High***

Q18. Assertion (A): Proteins are considered the primary and immediate source of energy during a highly explosive 100m sprint. **Reason (R):** Proteins are a macronutrient that yields 9 kcal of energy per gram. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) Both (A) and (R) are false. *Unit 5: Sports & Nutrition | [Practice] | Tag: High***

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

Q19. Enlist any two specific advantages of organizing a Knock-out tournament. *Unit 1: Management of Sporting Events* | [PYQ] | Tag: Very High**

Q20. Suggest any two corrective physical exercises or asanas for the lateral spinal deformity known as 'Scoliosis'. *Unit 2: Children & Women in Sports* | [PYQ] | Tag: Very High**

Q21. Mention any two major contraindications (medical conditions where it should be avoided) for performing *Bhujangasana* (Cobra Pose). *Unit 3: Yoga as Preventive Measure for Lifestyle Disease* | [PYQ] | Tag: High**

Q22. Briefly explain the primary role of a 'Physiotherapist' in assisting Children with Special Needs (CWSN). *Unit 4: Physical Education & Sports for CWSN* | [PYQ] | Tag: Very High**

Q23. Name any two test items included in the 'Johnson-Methney Test' of Motor Educability. *Unit 6: Test and Measurement in Sports* | [Practice] | Tag: Medium**

Q24. Differentiate briefly between the physiological terms 'Oxygen Intake' and 'Oxygen Uptake'. *Unit 7: Physiology & Injuries in Sport* | [PYQ] | Tag: High**

SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

Q25. Elaborate on the specific responsibilities of the 'Decoration and Ceremony Committee' and the 'Transport Committee' during the organization of a sports event. *Unit 1: Management of Sporting Events* | [PYQ] | Tag: High**

Q26. Define the 'Female Athlete Triad'. Explain in brief its core component 'Amenorrhea' and its primary physiological cause. *Unit 2: Children & Women in Sports* | [PYQ] | Tag: Very High**

Q27. Differentiate clearly between Simple Carbohydrates and Complex Carbohydrates by providing two points of distinction and one dietary example of each. *Unit 5: Sports & Nutrition* | [PYQ] | Tag: Very High**

Q28. Detail the purpose, equipment required, and the correct administration procedure of the 'Sit and Reach Test'. *Unit 6: Test and Measurement in Sports* | [PYQ] | Tag: Very High**

Q29. Regular participation in endurance sports delays the natural decline of bodily functions. Highlight any three physiological changes that occur in the human cardiovascular system specifically due to aging. *Unit 7: Physiology & Injuries in Sport* | [PYQ] | Tag: High**

Q30. State Newton's Third Law of Motion. Explain its practical application in sports by providing one distinct example each from Swimming and High Jump. *Unit 8: Biomechanics and Sports* | [PYQ] | Tag: Very High**

SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

Q31. A Physical Education teacher decides to organize an inter-class basketball tournament on a single league basis using the 'Staircase Method' to ensure every class gets to play against each other. A total of 12 classes (teams) have entered the competition. The school sports captain helps calculate the fixtures to avoid scheduling conflicts. **(i)** What will be the total number of matches played in this tournament? **(ii)** Calculate the total number of rounds that will be required to complete this fixture. **(iii)** In the staircase method of drawing league fixtures, are any 'Byes' given to the teams? (Yes/No) **(iv)** Write the mathematical formula used to calculate the number of matches if it were a *Double League* tournament. **OR** State the primary visual advantage of using the Staircase method for drawing fixtures. *Unit 1: Management of Sporting Events | [Practice] | Tag: Very High***

Q32. A professional weightlifter approaches a sports nutritionist to prepare for an upcoming national championship. The nutritionist designs a specific diet emphasizing high protein intake, moderate complex carbohydrates, and plenty of hydration. The nutritionist strongly advises him against using illegal steroids and educates him about 'Food Myths' that athletes blindly follow. **(i)** Physiologically, why does a weightlifter require a significantly higher intake of dietary protein compared to a normal person? **(ii)** Name the essential macro-mineral that the weightlifter must consume abundantly to ensure strong bone density and efficient muscle contractions. **(iii)** State any one common 'Food Myth' related to sports nutrition or dieting. **(iv)** Carbohydrates are the primary fuel for athletes. How many calories of energy does 1 gram of carbohydrate yield? **OR** Define the term 'Food Intolerance'. *Unit 5: Sports & Nutrition | [PYQ] | Tag: High***

Q33. During a fierce inter-school football final, player A is struck forcefully on the thigh by an opponent's knee, resulting in a dark blue swelling without the skin tearing. A few minutes later, player B awkwardly twists his ankle while landing after a header, tearing his ligament and collapsing in immense pain. The medical team rushes to the field. **(i)** The dark blue swelling on Player A's thigh, caused by ruptured internal blood vessels, is medically classified as a . **(ii)** The severe twisting injury that resulted in a torn ligament in Player B's ankle is termed as a . **(iii)** Both of these injuries are broadly categorized under which type of sports injuries? (Bone / Joint / Soft Tissue) **(iv)** The medical team immediately applies the **PRICE** protocol to Player B's ankle. What does the letter 'R' stand for in this acronym? **OR** Define the soft tissue injury known as an 'Abrasion'. *Unit 7: Physiology & Injuries in Sport | [PYQ] | Tag: Very High***

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 5)

Word limit: 200-300 words.

Q34. Define 'Obesity' as a major lifestyle disease. Name any four Yoga Asanas prescribed in the CBSE syllabus for the management of Obesity. Furthermore, thoroughly describe the step-by-step procedure and any two physiological benefits of practicing *Vajrasana*. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | [PYQ] | Tag: Very High***

Q35. What do you understand by the concept of 'Inclusion' in physical education? Elaborate in detail on any four practical strategies or modifications that a Physical Education teacher can implement to make sports activities accessible and enjoyable for Children With Special Needs (CWSN). *Unit 4: Physical Education & Sports for CWSN | [PYQ] | Tag: Very High***

Q36. The Harvard Step Test is a classic measure of cardiovascular fitness. Detail the complete administration of the Harvard Step Test. Your answer must include the required equipment (with exact specifications), the testing procedure, and the short formula used to calculate the athlete's Fitness Index. *Unit 6: Test and Measurement in Sports | [PYQ] | Tag: Very High***

Q37. Define 'Flexibility' as a component of physical fitness. Discuss comprehensively any three distinct physiological factors (such as joint structure, age, and muscle stretchability) that determine the level of flexibility in an athlete. Additionally, briefly explain the concept of 'Dynamic Equilibrium' with a suitable example from sports. *Unit 7 & 8: Physiology & Injuries / Biomechanics | [PYQ 2024] | Tag: High***

Answer Key

Q1. (B) 24 (Formula: $N - 1 = 25 - 1 = 24$). **Q2.** (A) Pes Planus **Q3.** (B) Ushtrasana **Q4.** (C) Paralympic Games **Q5.** (D) Vitamin D **Q6.** (B) Motor educability **Q7.** (C) Maximum Oxygen Uptake ($\dot{V}O_2$ max) **Q8.** (B) Opposite **Q9.** (B) League Tournament **Q10.** (C) Oligomenorrhea **Q11.** (B) Nadi Shodhana **Q12.** (C) Divisioning **Q13.** (D) Starch **Q14.** (B) 5 lbs **Q15.** (D) Dislocation **Q16.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q17.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q18.** (D) Both (A) and (R) are false. (Carbohydrates are the primary source; proteins yield 4 kcal/g, not 9).

Q19. 1. Highly economical as the tournament gets completed in less time requiring fewer resources. 2. Enhances the competitive standard as teams know a single defeat leads to elimination. **Q20.** 1. Hanging from a horizontal bar to stretch and decompress the spine. 2. Practicing *Trikonasana* (Triangle pose) or *Ardh Chakrasana* by bending in the opposite direction of the spinal curve. **Q21.** 1. Individuals suffering from severe hernia or recent abdominal surgeries should avoid it. 2. Pregnant women or individuals with extreme lower back conditions (like spondylitis) must not perform it. **Q22.** A Physiotherapist evaluates and treats CWSN using physical methods (exercises, massages, heat therapy) to improve their gross motor skills, joint mobility, muscle strength, and overall physical rehabilitation. **Q23.** 1. Front Roll. 2. Back Roll. (Also: Jumping Half-Turn, Jumping Full-Turn). **Q24. Oxygen Intake:** The amount of oxygen taken into the lungs from the atmosphere in one breath. **Oxygen Uptake:** The amount of oxygen actually absorbed from the blood and consumed by the working muscle cells for energy production.

Q25. Decoration and Ceremony Committee: Responsible for the aesthetic decoration of the sports venue/stadium, making arrangements for the opening and closing ceremonies, and organizing medals/trophies for the prize distribution. **Transport Committee:** Responsible for organizing safe and timely travel logistics (buses, cars) for athletes and officials from their

accommodation to the sports venue and back. **Q26. The Female Athlete Triad** is a syndrome consisting of three interrelated conditions—Eating Disorders, Amenorrhea, and Osteoporosis—stemming from low energy availability. **Amenorrhea** is the total absence of menstrual periods for three or more consecutive months. Its primary physiological cause is severe caloric deficit combined with extreme physical stress, causing the body's estrogen production to shut down. **Q27. Simple Carbohydrates:** 1. Consist of single/double sugar molecules, breaking down rapidly to provide immediate energy. 2. Taste sweet and spike blood sugar quickly. *Example:* Table sugar (Sucrose), glucose, honey. **Complex Carbohydrates:** 1. Consist of long chains of sugar molecules, breaking down slowly to provide sustained, long-lasting energy. 2. Generally not sweet to the taste. *Example:* Starch, whole wheat, potatoes. **Q28. Purpose:** To measure the flexibility of the lower back and hamstring muscles. **Equipment:** A specially constructed wooden 'Sit and Reach' box with a measurement scale extending over the top. **Procedure:** The athlete sits barefoot with legs fully extended and feet flat against the box. Keeping the knees straight, the athlete reaches forward with both hands along the measuring scale as far as possible and holds the stretch for 2 seconds. The maximum distance reached is recorded. **Q29. 1. Decreased Cardiac Output:** The heart muscle weakens, reducing the volume of blood pumped per minute. 2. **Decreased Maximum Heart Rate:** The heart's ability to beat rapidly at peak exertion drops naturally with age. 3. **Hardening of Blood Vessels:** Arteries lose their elasticity (atherosclerosis), leading to increased resting blood pressure and reduced blood flow to muscles. **Q30. Newton's Third Law (Law of Action & Reaction):** For every action, there is an equal and opposite reaction. **Swimming:** An athlete pushes the water backward using their hands and feet (Action), and the water exerts an equal force to push the swimmer forward (Reaction). **High Jump:** The jumper thrusts their take-off foot forcefully *downward* against the ground (Action), and the ground provides an equal reactive thrust *upward*, propelling the athlete into the air (Reaction).

Q31. (i) Matches = $12(11)/2 = 132/2 = 66$ matches. (ii) Rounds = $N - 1 = 12 - 1 = 11$ rounds. (iii) No. (iv) Double League Formula: $N(N - 1)$. **OR** Advantage: It is visually simple to draw and understand, and completely eliminates the need to calculate or allocate 'byes' regardless of whether teams are odd or even. **Q32.** (i) Weightlifters undergo intense resistance training that creates micro-tears in muscle fibers; high dietary protein is strictly required to repair these tissues and promote muscle hypertrophy (growth). (ii) Calcium. (iii) "Carbohydrates make you fat" or "Starving helps in quick weight loss." (iv) 4 kcal per gram. **OR** Food intolerance is a digestive system issue where the body lacks the specific enzymes needed to break down certain food components (like lactose), causing bloating, nausea, or stomach cramps without an immune response. **Q33.** (i) Contusion. (ii) Sprain. (iii) Soft Tissue Injuries. (iv) 'R' stands for **Rest** (immediately stop playing to prevent further damage). **OR** An abrasion is a superficial soft tissue injury caused by friction or scraping of the skin against a rough surface (e.g., sliding on turf), stripping away the epidermis layer.

Q34. Obesity is an unhealthy accumulation of excess body fat resulting in a BMI of 30 or above, posing severe health risks like diabetes and heart disease. **Asanas:** Vajrasana, Hastotanasana, Trikonasana, Ardha Matsyendrasana (Any four from syllabus). **Vajrasana Procedure:** Sit on the floor with legs stretched straight. Fold the right leg and place the foot

under the right hip. Fold the left leg and place it under the left hip. Keep the heels slightly apart and toes together. Sit straight keeping the spine, neck, and head vertically aligned. Rest the palms on the knees and breathe deeply. **Benefits:** 1. It is the only asana that can be done immediately after meals, drastically improving digestion and preventing acidity. 2. It strengthens the pelvic muscles and relieves lower back pain/sciatica. **Q35. Inclusion** means providing equal opportunities to CWSN to participate fully in mainstream educational and physical activities without discrimination. **Strategies for Inclusion:** 1. **Adapted Equipment:** Modify the sporting gear to suit the disability, such as using lighter racquets, larger softballs, or auditory balls equipped with bells for visually impaired children. 2. **Rule Modifications:** Altering the rules of the game to ensure success, such as allowing an extra bounce in tennis, extra time to complete a race, or shortening the field dimensions. 3. **Accessible Environment:** Ensuring the physical infrastructure is barrier-free by installing wheelchair ramps, smooth play surfaces, and removing loud background noises for children with sensory processing disorders. 4. **Peer Buddy System:** Pairing a mainstream student (buddy) with a CWSN athlete during play. The buddy provides physical assistance, emotional encouragement, and helps the special child integrate socially with the team. **Q36. Equipment:** A wooden stepping bench exactly 20 inches high for men and 16 inches high for women, a stopwatch, and a metronome (or cadence tape). **Procedure:** 1. The athlete steps up and down on the bench at a constant pace of 30 steps per minute for exactly 5 minutes (or until exhaustion). 2. Immediately after finishing, the athlete sits down to recover. 3. The tester counts the athlete's heartbeats (pulse) during the recovery period at specific intervals: from 1 to 1.5 minutes, 2 to 2.5 minutes, and 3 to 3.5 minutes. **Fitness Index Formula (Short Method):** $FI = \frac{100 \times \text{Test Duration in Seconds}}{5.5 \times \text{Pulse count between 1 to 1.5 minutes}}$ **Q37. Flexibility** is the maximum range of motion possible around a joint. **Physiological Factors:** 1. **Joint Structure:** The anatomical build of a joint dictates flexibility. Ball-and-socket joints (like the shoulder) inherently permit a much wider range of motion in all directions compared to hinge joints (like the knee). 2. **Age and Gender:** Flexibility naturally decreases with age due to tendons losing elasticity. Females generally possess greater flexibility than males due to anatomical pelvic differences and hormones (relaxin/estrogen). 3. **Stretchability of Muscles:** The inherent elasticity of the muscle tissues and their surrounding ligaments heavily determines how far a joint can stretch without tearing. **Dynamic Equilibrium:** It is the ability to maintain the body's center of gravity and balance while in active, uniform motion. *Example:* A gymnast executing a cartwheel or a cyclist safely navigating a sharp velodrome curve without falling over.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.